When Survival Improves But Quality of Life Does Not: A Model-Based Meta-Analysis of Immune Checkpoint Inhibitors

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1. Model Source Codes "test8 – Final"

The provided folder contains the complete Monolix project files and exported model outputs. The project is organized around a Monolix run named test8, with the main source code files located at the top level and model outputs exported into the test8/ results directory.

At the top level, the folder includes:

- **test8.mlxtran**: the main Monolix project file defining the dataset, covariates, structural model, individual parameter model, residual error model, estimation tasks, and plot-generation settings.
- **test8.txt**: the longitudinal model code file called by test8.mlxtran. This file contains the mathematical model used to predict transformed quality-of-life (QoL) outcomes.
- **test8.mlxproperties**: Monolix plotting and visualization settings, including settings for the visual predictive check.

The exported result folders include:

- **Bootstrap/**: bootstrap parameter summaries and estimates.
- **ChartsData/**: source data for diagnostic plots, including individual fits, observed vs predicted values, residuals, VPC, parameter distributions, and random effects.
- **ChartsFigures/**: exported diagnostic figures as PNG files.
- **FisherInformation/**: covariance and correlation estimates from the Fisher Information Matrix.
- **IndividualParameters/**: estimated and simulated individual parameters and random effects.
- **LogLikelihood/**: OFV, AIC, BIC, and BICc.
- **Tests/**: statistical tests for fixed effects, residual normality/symmetry, random effects, and covariate relationships.
- **summary.txt**: comprehensive Monolix run summary.
- **populationParameters.txt** and **populationParametersByGroups.txt**: population parameter estimates and group-specific estimates.
- **predictions.txt** and **tables.txt**: model predictions and exported model table outputs.

The model uses a transformed QoL endpoint, incorporates baseline QoL and sample-size-related regressors, and estimates an Emax-type longitudinal trajectory with slope and treatment-group effects. The main result suggests that the control group has a lower slope parameter than the ICI group, while the model also accounts for between-study and between-treatment-arm variability in Emax. The exported Monolix outputs include full population estimates, individual estimates, uncertainty estimates, bootstrap summaries, diagnostic tests, visual predictive check data, and diagnostic figures.

2. Study Description – "Summary of studies and data"

This dataset summarizes study-level characteristics and extracted outcomes from published clinical trials included in the MBMA analysis. It integrates information on study identifiers, treatment types, disease indications, QoL reporting, survival outcomes, and data availability. The dataset is used to track data completeness, assess between-group differences, and support integration of QoL and survival data across studies.

Variable	Definition
STID	Study identifier
Study	First author and study reference
PubMed ID	Unique identifier for the publication in PubMed
Clinical trial ID	ClinicalTrials.gov identifier
ICI type	Type of immune checkpoint inhibitor (ICI) treatment
Disease type	Cancer type investigated in the study
Reported between-group difference (PRO)	Whether clinically meaningful or statistically significant QoL difference between ICI and control arm was reported
QoL Note	Additional notes on QoL
Number of data points (ICI)	Number of QoL data points extracted for treatment arm
Number of data points (Control)	Number of QoL data points extracted for control arm
Baseline sample size (ICI)	Sample size of treatment arm at baseline
Baseline sample size (Control)	Sample size of control arm at baseline

Survival Benefits	Whether survival benefit was observed
Survival Benefits added with KM data results	Whether survival conclusion was supplemented using KM curve extraction
PUBMED ID OS	PubMed ID for survival outcome reference
PUBMED ID OS 2	Secondary PubMed ID for survival data
Note	Additional comments
Final KM collected	Whether KM data were successfully extracted

3. Source data of quality of life – "QoL data.csv"

This dataset summarizes individual study- and arm- level longitudinal quality of life (QoL) data extracted from clinical trials using instruments such as EORTC QLQ-C30. The dataset is used to build nonlinear mixed effect model.

Variable	Definition	Unit
ID	Unique identifier of study	N/A
STID	Study identifier matching "Summary of studies and data"	N/A
OID	Unique identifier of arm	N/A
PUBMEDID	PubMed ID of the publication source	N/A
NCT	ClinicalTrials.gov identifier for the study	N/A
DISEASE	Disease or indication studied in the trial	N/A
DRUG	Treatment drug/control in the arm	N/A
DRUG2	Treatment category	N/A
TIMEWK	Time point of measurement	Weeks
TransDV	Logit transformed of QoL scores	
RESPONSEFINAL	QoL score at each time point	point
NTOTAL	Total number of participants in the study	N/A
NOC	Number of participant contributing to that time point (arm-level)	N/A
BSL	Baseline QoL score	point
AGE	Mean or median age of patients	Years
Male	Proportion or percentage of male patients	%

ECOG0	Proportion of patients with ECOG performance status = 0	%
ECOG1	Proportion of patients with ECOG performance status = 1	%
DSI	Proportion of patients with Disease Stage I	%
DSII	Proportion of patients with Disease Stage II	%
DSIII	Proportion of patients with Disease Stage III	%
DSIV	Proportion of patients with Disease Stage IV	%

4. Source data of overall survival – "overall_survival.xlsx"

This dataset contains digitized Kaplan-Meier curves data.

Variable	Definition	Unit
OID	Unique arm identifier matching "QoL data.csv"	N/A
Time	Time to event or censoring	Months
OS	Overall survival probability	

5. Source data of paired study arms – "OID_control_ICI.xlsx"

This table maps study IDs (STID) to arm IDs (OID) for treatment and control groups, enabling consistent pairing of comparisons within each study.

Variable	Definition
STID	Study identifier matching "Summary of studies and data"
ICI_OID	Unique arm identifier matching "QoL data.csv" for immune checkpoint inhibitor (ICI) treatment within each study
control_OID	Unique arm identifier matching "QoL data.csv" for control group within each study
Note	Additional information for studies with special designs

6. Source data of individual survival data – "IPD_data_6.7.xlsx"

This dataset contains individual-level survival data reconstructed from published Kaplan–Meier curves. It includes time-to-event information and censoring status for each observation, grouped by arm identifier (OID). The dataset is used to model and compare survival outcomes across treatment groups.

Variable	Definition	Unit
OID	Unique arm identifier matching "QoL data.csv"	N/A
Time	Time to event or censoring	Months
Censored	Event indicator (1 = censored, 0 = event occurred)	Binary